

Ans1. Integer(int) :-

- An integer is whole number without decimal points.
- It is mainly used indexing , counting and mathematical calculations.
- Example :-
Age = 20

Float(float):-

- Float is a number that contains a decimal point.
- Float is mainly used for scientific calculations , financial data , measurements , etc.
- Example:-

pi = 3.14159

String(str):-

- String is sequence of characters enclosed in double or single quotes.
- It is used to store text such as names , addresses , messages , etc.
- Example:-
name = "Apoorva"

Boolean(bool):-

- It is a data type that has 2 values : True/False.
- It is used in decision-making and conditional statements.
- Example:-
x = 10
y = 20
print(x<y)
#Output : True

Ans4. Five mainly used string methods in python are :-

1. upper():-

- Converts all characters into uppercase of a given string.
- Example :-
name = "apoorva"
print(name.upper())

2. lower():-

- Converts all characters into lowercase of a given string.
- Example :-

name = "apoorva"
print(name.lower())

3. capitalize():-

- This converts the first character to uppercase and the remaining characters to lowercase.
- Example:-
name = "apoorva"
print(name.capitalize())

4. replace():-

- This method replaces a specified character or word with another.
- Example:-
intro = "I love Java"
print(intro.replace("Java", "Python"))

5. strip():-

- This method removes extra spaces from the beginning and end of a string.
- Example:-
text = " Python "
print(text.strip())

Ans7. AI is a technology that enables machines and computers to perform tasks that require human intelligence.

Importance of AI:-

- Efficiency and Automation: AI systems streamline repetitive tasks, allowing businesses to operate 24/7 and increasing overall productivity.
- Data-Driven Insights: AI can analyze vast amounts of data in seconds, uncovering patterns that humans might miss, which leads to highly accurate predictions and smarter decision-making.
- Continuous Improvement: Through machine learning, AI algorithms get smarter over time, constantly improving their accuracy and performance without human intervention.
- Innovation: AI solves highly complex scientific problems, from modeling climate change to accelerating pharmaceutical drug discovery.

Real-Life Applications:-

- Healthcare: AI powers predictive diagnostics (e.g., detecting tumors in medical imaging), personalizes treatment plans, and assists in robotic surgeries. [[1](#), [2](#), [3](#), [4](#), [5](#)]
- Transportation: Autonomous self-driving vehicles (like Tesla or Waymo) and intelligent traffic management systems rely heavily on AI computer vision and sensors.
- Customer Service: Virtual assistants and smart chatbots handle inquiries, troubleshoot issues, and provide instant support across retail and banking platforms.
- Entertainment & Media: Recommendation engines on streaming platforms like Netflix and Spotify use AI to analyze your preferences and curate personalized content feeds. [[1](#), [2](#), [3](#)]
- Finance: AI algorithms monitor transactions in real-time to detect fraudulent activities and power automated high-frequency trading. [[1](#), [2](#), [3](#)]
- Everyday Convenience: Smart home devices (e.g., Amazon Alexa, Google Assistant) and mobile apps like navigation (Google Maps) use AI for voice recognition, route optimization, and contextual awareness.

Ans8. 1. ChatGPT

Yes, it is AI.

ASSIGNMENT -1

ChatGPT can understand our questions and give answers like a human. It uses AI to learn from a large amount of data and help users.

2. Google Maps Route Prediction

Yes, it is AI.

Google Maps uses AI to check traffic and road conditions. It then suggests the best route and estimates the travel time.

3. Calculator

No, it is not AI.

A calculator only follows the instructions programmed into it. It cannot learn, think, or make decisions on its own.

4. Netflix Recommendations

Yes, it is AI.

Netflix uses AI to suggest movies and shows based on what a user has watched before.

5. Voice Assistants (Alexa/Siri)

Yes, they are AI.

Alexa and Siri can understand voice commands and answer questions. They use AI to interact with users and perform tasks.